1 OUTCOME-BASED CURRICULUM (120)

Total Marks 120.00

1.1 Vision, Mission and Program Educational Objectives (PEOs) (35)

Total Marks 35.00

1.1.1 State the Vision and Mission of the Institute and the Department (5)

Institute Marks : 5.00

Vision of the institute	To be a leading institution in engineering education, fostering innovation, research, and ethical values to serve society.
Mission of the institute	• To provide quality education through effective teaching-learning processes. • To promote research, innovation, and entrepreneurship. • To collaborate with industry and society for sustainable development. • To develop competent and socially responsible professionals.
Vision of the Department	To generate competent professionals to become part of the Industry and Research Organizations at the National and International levels.

Mission of the Department

Mission No.	Mission Statements
M1	To train the students to have in-depth knowledge of the subjects in the field of Computer Science and Engineering
M2	To train the students with leadership qualities, teamwork skills, commitment and ethics thereby making them develop confidence for R & D activities and for placement in multinational and national.

1.1.2 State PEOs of the Program (5)

Institute Marks : 5.00

PEO No.	Program Educational Objectives Statements

PEO1	Graduates will have successful technical or professional careers, including supportive and leadership roles on multidisciplinary teams.
PEO2	Graduates will be able to acquire, use, and develop skills required for effective professional practices.
PEO3	Graduates will be able to attain a holistic education that is an essential prerequisite for being a responsible member of society.

1.1.3 Process of Defining Vision, Mission and PEOs (10)

Institute Marks : 10.00

Process of Defining Department Vision and Mission of the Department:

The formulation and refinement of the Department Vision and Mission are carried out through a structured and participative process in alignment with the guidelines of the National Board of Accreditation.

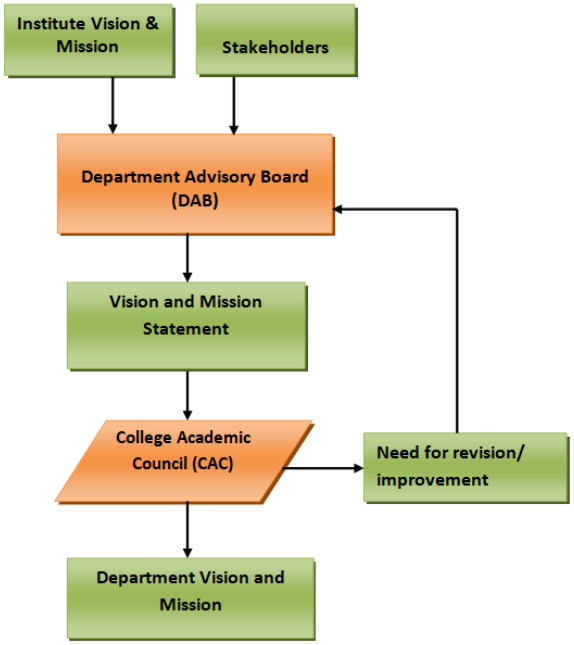
Initially, the **Institute Vision and Mission** serve as the primary reference framework, ensuring that the department goals are aligned with the overall institutional objectives. Simultaneously, inputs are collected from various **stakeholders**, including faculty, students, alumni, and industry experts, to incorporate diverse perspectives and current industry requirements.

These inputs are presented to the Department Advisory Board, which plays a crucial role in consolidating and analyzing the information. The DAB reviews the institutional vision along with stakeholder expectations and formulates a draft **Vision and Mission statement** for the department.

The draft is then forwarded to the **College Academic Council (CAC)** for further evaluation and approval. The CAC examines the proposed statements to ensure academic rigor, relevance, and compliance with regulatory standards.

During the review process, if any gaps or shortcomings are identified, a **need for revision or improvement** is recognized. The feedback is communicated back to the Department Advisory Board, and necessary modifications are incorporated. This establishes a feedback loop ensuring continuous refinement.

After incorporating all suggestions and obtaining necessary approvals, the finalized **Department Vision and Mission** are formally adopted and disseminated.



Process of Defining Program Educational Objectives:

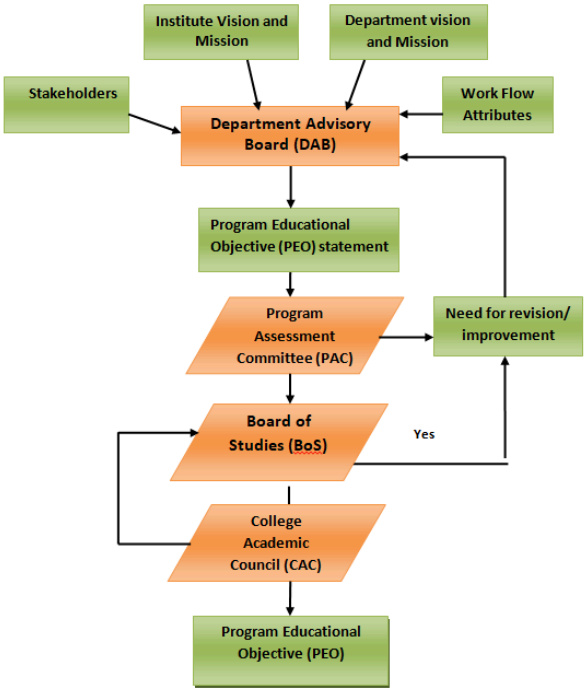
The process of establishing the mapping between Program Educational Objectives (PEOs) and Mission statements is carried out in a systematic and structured manner in alignment with Outcome-Based Education (OBE) principles and the guidelines of the National Board of Accreditation.

Initially, the Vision and Mission statements of the Institute and the Department are thoroughly reviewed to ensure that the PEOs are consistent with the overall educational goals. The PEOs are formulated based on inputs collected from key stakeholders such as employers, alumni, faculty members, and students, reflecting the expected achievements of graduates within three to five years after completion of the program.

The Department Mission is then decomposed into distinct components (M1, M2,) representing core areas such as technical competence, research and innovation, ethical responsibility, and lifelong learning.

A correlation matrix is developed by mapping each PEO with the Mission components using a predefined scale of 1 (Low), 2 (Moderate), and 3 (High), indicating the level of alignment. The mapping is carefully justified to ensure logical consistency and relevance. This matrix is subsequently reviewed and validated by the Department Advisory Board, incorporating suggestions from academic and industry experts.

The finalized mapping is approved by statutory bodies such as the Board of Studies (BoS) and CAC. The approved PEO–Mission mapping is documented and disseminated through various platforms including the department website, course files, and laboratory manuals. The entire process is periodically reviewed based on feedback from stakeholders and attainment analysis to ensure continuous improvement and alignment with evolving industry and societal needs.



1.1.4 Dissemination of Vision, Mission and PEOs (5) | Institute Marks : 5.00

Dissemination of Vision, Mission and PEOs

The mission and vision are published through display in the following

- Institute and Department website
- College Library and Department library
- Lab Manuals
- Faculty Room,Class Room and laboratories
- LMS Portal
- Syllabus Books
- Course Files
- Notice Boards

Process of dissemination among stakeholders

- Vision, Mission and PEOs are disseminated in the college website for internal and external stakeholders.

- During orientation day the parents and students are informed about the vision and mission of the institute and the department. This helps the institution to disseminate the information to students and parents.
- Vision and Mission is disseminated to all the stakeholders of the programs through faculty meetings, student awareness workshops, student induction programs, and parent meetings.
- PEOs can be disseminated among stakeholders through electronic mails and also in various national and international events organized in the institute.
- During meetings the principal apprises the faculty and staff members regarding the vision and mission of the institute. Head of the department acquaints the vision, mission and PEOs of the department to the Faculty in the department meeting.
- Training and placement officer informs regarding vision and mission of the institute and department to the industry personnel during their visits to industry. As a result the industry personnel are made aware of the vision and mission of the institute.

Dissemination among Stakeholders:

Dissemination and publishing of Vision, Mission and PEOs among the stakeholders summarized in the following table:

S.No	Location	Institute Vision	Institute Mission	Department Vision	Department Mission	PEO&PSO
1	Institute Website / Departmental Webpage	✓	✓	✓	✓	✓
2	Department News Letter & Notice Board	✓	✓	✓	✓	✓
3	Course file	✓	✓	✓	✓	✓
4	Lab Manual	✓	✓	✓	✓	✓
5	Conference / workshop Brochures	✓	✓	✓	✓	

The Internal Stakeholders are

1. Faculty Members
2. Students
3. Representatives from the Management

The External Stakeholders are

1. Industry Experts
2. Body of Governance
3. Alumni
4. Recruiters
5. Parents

6. Society

7. Funding Agencies

8. Educators

1.1.5 Mapping of PEOs with Mission (10) | Institute Marks : 10.00

PEO Statements	M1	M2
Graduates will have successful technical or professional careers, including supportive and leadership roles on multidisciplinary teams.	2	3
Graduates will be able to acquire, use, and develop skills required for effective professional practices.	3	2
Graduates will be able to attain a holistic education that is an essential prerequisite for being a responsible member of society.	3	2

Justification and rationale for mapping of PEOs and Mission Statements

PEO1 → M1 = 2

PEO1 focuses on fundamental knowledge and problem-solving. M1 helps students gain in-depth knowledge in Computer Science and Engineering. This supports PEO1 well, but it mainly focuses on subject knowledge. Hence, the relation is moderate.

PEO1 → M2 = 3

M2 develops leadership, teamwork, commitment, and confidence. These qualities help students apply their knowledge in real-life situations and work effectively in multidisciplinary teams. Therefore, PEO1 has a strong mapping with M2.

PEO2 → M1 = 3

PEO2 includes problem-solving and professional practice. M1 gives students strong technical knowledge and depth in the core subjects. This helps them analyze problems and find suitable solutions. So, the mapping between PEO2 and M1 is high.

PEO2 → M2 = 2

M2 supports professional growth through leadership, teamwork, and ethics. These qualities are useful for professional practice, but they support PEO2 indirectly when compared with the technical strength provided by M1. Hence, the mapping is moderate.

PEO3 → M1 = 3

PEO3 focuses on holistic education and becoming a responsible member of society. M1 contributes by building strong academic understanding, analytical thinking, and disciplined learning. These are important parts of holistic development. Therefore, PEO3 is given a high mapping with M1.

PEO3 → M2 = 2

M2 supports ethics, teamwork, and commitment, which are useful for social responsibility. However, in this mapping, the department appears to give slightly more weight to academic and intellectual growth under M1 for achieving PEO3. Hence, the mapping with M2 is moderate.

1.2 Curriculum Structure and Features (30) | Total Marks 30.00

1.2.1 State the Process for Developing/Revising the Program Curriculum (10) | Institute Marks : 10.00

Process for Developing/Revising the Program Curriculum

Anurag Engineering College (ANRK) is affiliated with Jawaharlal Nehru Technological University Hyderabad and functions as an autonomous institution. Leveraging its autonomous status, the institution systematically designs and develops its curriculum in alignment with the guidelines of the affiliating university, AICTE Model Curriculum, and the Washington Accord Knowledge and Attitude Profiles (WKs).

The curriculum framework is structured to ensure **Outcome-Based Education (OBE)**, focusing on the attainment of **Program Outcomes (POs), Program Specific Outcomes (PSOs), and Course Outcomes (COs)**, thereby enhancing students' knowledge, skills, and professional competencies to meet industry requirements upon graduation.

The institution follows a **well-defined and participatory curriculum revision process**, wherein the syllabus is reviewed and updated once every three years. The revision process involves active consultation with key stakeholders, including:

- Industry experts
- Academic experts
- Alumni
- Employers
- Parents

Feedback collected from these stakeholders is systematically analyzed and incorporated to ensure relevance, quality, and continuous improvement in curriculum design.

Currently, the institution is implementing multiple regulations to ensure smooth academic progression:

- **R25 Regulation** – Applicable for students admitted from the academic year 2026–27 onwards
- **R22 Regulation** – Applicable for students admitted during 2022–23 to 2025–26
- **R18 Regulation** – Applicable for students admitted during 2018–19 to 2021–22

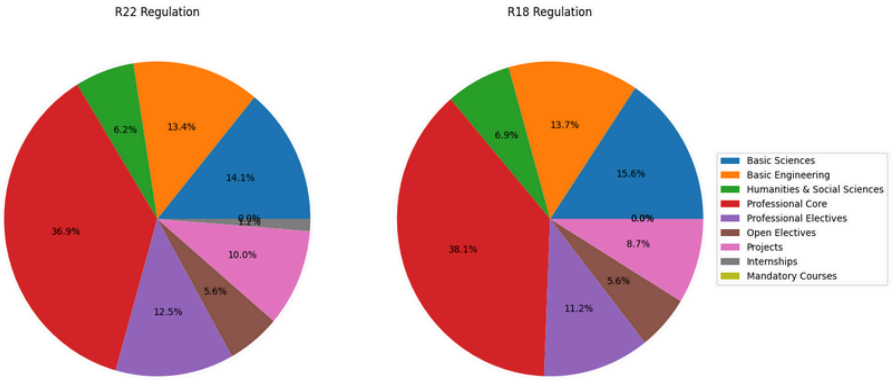
The following table presents the mapping of regulations with respect to the academic years and the corresponding batches of students.

Academic Year	I B.Tech	I B.Tech	II B.Tech	IV B.Tech
2025-26	R25	R22	R22	R22
2024-25	R22	R22	R22	R18
2023-24	R22	R22	R18	R18

The curriculum given by the university is a composition of subjects related to social sciences & humanities, basic sciences, engineering sciences, program core courses, program electives, open elective courses, project & seminar that make the students apply the learnt engineering knowledge to analyze and design solutions to complex problems with social consciousness and ethics. The

course modules include credit and non-credit courses and their percentage contribution to the Computer Science and Engineering program is given in Table

	R22 Regulation		R18 Regulation	
Curriculum Components	**No. of Courses**	**Curriculum Content (% of total number of credits of the program)**	**No. of Courses**	**Curriculum Content (% of total number of credits of the program)**
Basic Sciences	7	14.06	8	15.63
Basic Engineering	11	13.44	11	13.75
Humanities and Social Sciences	5	6.25	6	6.88
Professional Core	28	36.88	24	38.13
Professional Electives	7	12.50	6	11.25
Open Electives	3	5.63	3	5.63
Project(s)	3	10.00	4	8.75
Internships/ Industry-Oriented Mini-Project	1	1.25	0	0
Any other - Mandatory Courses	4	0.00	4	0
Total no.of Courses	69	100	66	100



Process for Developing/Revising the Program Curriculum ensures alignment with **global standards** (such as the Washington Accord), **national requirements** (such as NBA-defined POs, Credit Framework by UGC, India skill report), and the **departments Vision, Mission, PEOs, and PSOs**.The process is **collaborative**, involving **internal stakeholders** (faculty, students, and department leadership) and **external stakeholders** (industry experts, alumni, and academic experts). Below is a detailed step-by-step description of the process as it is currently followed:

1. Formation of the Department Advisory Board (DAB)

The DAB is formed with the following members:

- **Senior faculty members** from the department with diverse specializations (e.g., AI, cybersecurity, software engineering).

- **Representatives from Science and Humanities** to ensure foundational subjects align with engineering requirements.
- **Faculty from other core branches** to promote interdisciplinary learning.
- **Industry experts** and **alumni** to provide practical insights and ensure industry relevance.
- **Educators from various fields.**

The PAC is responsible for overseeing the entire curriculum development process, from input collection from DAB to final implementation.

2. Input Collection

The DAB gathers inputs from multiple sources to ensure the curriculum is comprehensive, globally aligned, and industry-relevant. The inputs include:

- **UGC Choice Based Credit System (CBCS) Framework and Credit Framework**: Ensures compliance with national higher education guidelines.
- **AICTE Model Curriculum**: Provides a benchmark for technical education standards in India.
- Aligns the curriculum with global best practices from **international universities**, incorporating research-oriented learning, modern teaching methodologies, and global accreditation standards.
- Integrates **Skill India initiatives** by embedding industry-relevant training, NSQF-aligned skill programs, and hands-on certification courses for enhanced employability.
- **NBA-defined Program Outcomes (POs)**: Ensures the curriculum aligns with the 12 graduate attributes defined by the NBA, such as engineering knowledge, problem analysis, design/development of solutions, and ethics.
- **Parent University Regulations and Curriculum**: Ensures compliance with university-level academic requirements.
- **Department Vision, Mission,POs, PEOs, and PSOs**: Ensures the curriculum supports the departments long-term goals and objectives.
- **Feedback from Stakeholders**: Collects insights from students, faculty, alumni, and industry experts to identify gaps and areas for improvement.

3. Draft Curriculum Preparation

Based on the collected inputs, the PAC prepares a **draft curriculum**.

The draft is designed to:

- Align with the **NBA- defined POs**.
- Incorporate **emerging technologies** and **industry trends**
- Promote **interdisciplinary learning** and **project-based education**.
- Include **value-added courses** on ethics, sustainability, and entrepreneurship.

The PAC is responsible for overseeing the entire curriculum development process, from input collection from DAB to final implementation.

4. Stakeholder Feedback

The draft curriculum is shared with **internal and external stakeholders** for feedback. The feedback collection process includes:

- **Workshops**: Organized with industry experts, academicians, and alumni to discuss the draft curriculum and suggest improvements.
- **Online Surveys**: Distributed to students, faculty, and alumni to gather their perspectives on the proposed changes.
- **Industry Consultations**: Direct interactions with industry partners to ensure the curriculum meets current and future industry needs.

5. Modification and Finalization

The PAC reviews the feedback and incorporates necessary modifications into the draft curriculum. Key considerations during this stage include:

- Ensuring alignment with **POs**.
- Addressing gaps identified by stakeholders.
- Balancing theoretical knowledge with practical skills.
- Incorporating **global best practices** and **emerging trends**.

Once the modifications are complete, the curriculum is finalized and prepared for approval from BOS.

6. Course Allocation and Content Preparation

After finalizing the curriculum, the PAC allocates courses to faculty members based on their expertise. Faculty members are responsible for:

- Preparing **detailed course contents**, including learning objectives, topics, and reference materials.
- Designing **assessment components** (e.g., exams, projects, presentations) to evaluate student learning outcomes.
- Ensuring that each course aligns with the overall program objectives and POs.

7. Board of Studies (BOS) Approval.

The finalized curriculum and course content are presented to the **Board of Studies (BOS)** by the BOS Chairman for approval. The BOS comprises:

- Senior faculty members from the department and other branches.
- Industry experts and academicians.
- Alumni representatives.

The BOS reviews the curriculum approves or provides feedback and suggestions for improvement. The PAC incorporates the feedback and prepares the final version for approval.

8. Ratification and Implementation

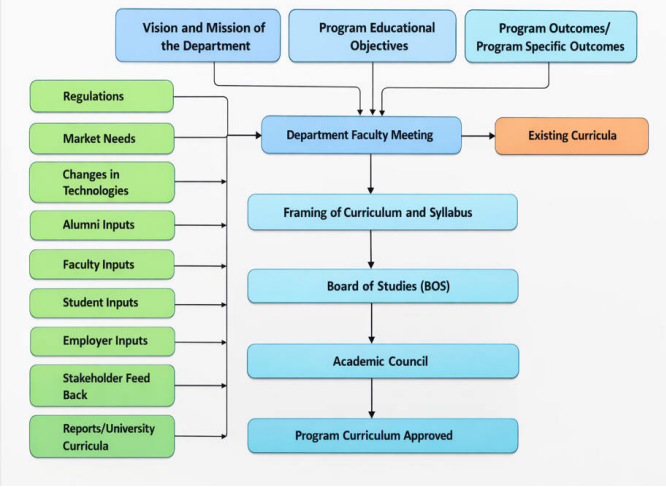
The final curriculum is submitted for ratification in the College Academic Council Meeting. Once ratified, the curriculum is:

- Published and disseminated to all stakeholders, including students, faculty, and industry partners.
- Implemented in the academic program, with faculty members delivering the courses as per the curriculum.

9. Continuous Improvement

The curriculum development process is **dynamic** and **iterative**, with mechanisms in place for continuous improvement:

- **Ongoing Feedback**: Suggestions for curriculum changes are encouraged from students, faculty, and other stakeholders throughout the academic year.
- **Immediate Actions**: Recommendations requiring urgent implementation are addressed and ratified in subsequent BOS and AC Meetings.
- **Periodic Revisions**: The curriculum is reviewed and revised every **2 years** to incorporate emerging trends, technological advancements, and stakeholder feedback.



1.2.2 Curriculum Structure (10) | Institute Marks : 10.00

Edit

ID	Course Code	Course Title	Classroom Instruction (CI) (in hours per semester)	Lab Instruction (LI) (in hours per	Term Work (TW) and Self Learning	Total no. of Hours per semester	Total Credits (C)*

			semester)			(SL) (TW+ SL) (in hours per semester)		(Total Hours/30)
			L	T	P	SL		
1	CS703PC	Linux Programming Lab	0	0	30	0	30	1.00
2	CS704PW	Mini project	0	0	0	60	60	2.00
3	CS705PW	Project Stage-I	0	0	0	120	120	4.00
4	CS852PE,CS853PE	Professional Elective-V	90	0	0	0	90	3.00
5	CS861PE,CS862PE	Professional Elective-VI	90	0	0	0	90	3.00
6	CE831OE,CE832OE	Open Elective-III	90	0	0	0	90	3.00
7	CS801PW	Seminar	0	0	0	60	60	2.00
8	CS802PW	Project Stage-II	0	0	0	180	180	6.00
9	MA101BS	Matrices and Calculus	90	30	0	0	120	4.00
10	CH102BS	Engineering Chemistry	90	30	0	0	120	4.00
11	CS103ES	Programming for Problem Solving	90	0	0	0	90	3.00
12	EE104ES	Basic Electrical Engineering	60	30	0	0	90	3.00
13	EG105ES	Computer Aided Engineering Graphics	30	0	60	0	90	3.00
14	CS106ES	Elements of Computer Science and Engineering	0	0	30	0	30	1.00
15	CH107BS	Engineering Chemistry Laboratory	0	0	30	0	30	1.00
16	CS109ES	Programming for Problem Solving laboratory	0	0	30	0	30	1.00
17	EE108ES	Basic Electrical Engineering Laboratory	0	0	30	0	30	1.00
18	HS110MC	Constitution of India	0	0	0	0	0	0.00
19	MA201BS	ODE and Vector Calculus	90	30	0	0	120	4.00
20	AP202BS	Applied Physics	90	30	0	0	120	4.00
21	ME203ES	Engineering Workshop	0	30	45	0	75	2.50
22	EN204HS	English for Skill Enhancement	60	0	0	0	60	2.00

23	EC205ES	Electronic Devices and Circuits	60	0	0	0	60	2.00
24	AP206BS	Applied Physics Laboratory	0	0	45	0	45	1.50
25	CS207ES	Python Programming Laboratory	0	30	30	0	60	2.00
26	EN208HS	English Language and Communication Skills Laboratory	0	0	30	0	30	1.00
27	CS209ES	IT Workshop	0	0	30	0	30	1.00
28	ES210MC	Environmental Science	0	0	0	0	0	0.00
29	PS301BS	Probability and Statistics	90	30	0	0	120	4.00
30	DE302ES	Digital Electronics	90	0	0	0	90	3.00
31	CS303PC	Data Structures	90	0	0	0	90	3.00
32	CS304PC	Computer Organization and Architecture	90	0	0	0	90	3.00
33	CS305PC	Object Oriented Programming through Java	90	0	0	0	90	3.00
34	CS306PC	Data Structures Laboratory	0	0	45	0	45	1.50
35	CS307PC	Object Oriented Programming through Java Laboratory	0	0	45	0	45	1.50
36	HS309MC	Gender Sensitization	0	0	0	0	0	0.00
37	CS308PC	Skill Development Course(Data visualization-R Programming / Power BI)	0	0	30	0	30	1.00
38	MB401HS	Business Economics & Financial Analysis	90	0	0	0	90	3.00
39	CS402PC	Discrete Mathematics	90	0	0	0	90	3.00
40	CS403PC	Operating Systems	90	0	0	0	90	3.00
41	CS404PC	Database Management Systems	90	0	0	0	90	3.00
42	CS405PC	Software Engineering	90	0	0	0	90	3.00
43	CS406PC	Operating Systems Laboratory	0	0	30	0	30	1.00
44	CS407PC	Database Management Systems Laboratory	0	0	30	0	30	1.00
45	CS408PW	Real-time Research Project/ Societal Related Project	0	0	60	0	60	2.00
46	HS411MC	Intellectual Property Rights	0	0	0	0	0	0.00

47	CS409PC	Skill Development Course(Node JS/ React JS/ Django)	0	0	30	0	30	1.00
48	CS501PC	Design and Analysis of Algorithms	90	30	0	0	120	4.00
49	CS502PC	Computer Networks	90	0	0	0	90	3.00
50	CS503PC	DevOps	90	0	0	0	90	3.00
51	CS513PE,CS515PE	Professional Elective-I	90	0	0	0	90	3.00
52	CS523PE,CS524PE	Professional Elective-II	90	0	0	0	90	3.00
53	CS504PC	Computer Networks Laboratory	0	0	30	0	30	1.00
54	CS505PC	DevOps Laboratory	0	0	30	0	30	1.00
55	AE506HS	Advanced English Communication Skills Laboratory	0	0	30	0	30	1.00
56	CS507PC	Skill Development Course(UI design- Flutter)	0	0	30	0	30	1.00
57	CS601PC	Machine Learning	90	0	0	0	90	3.00
58	CS602PC	Formal Languages and Automata Theory	90	0	0	0	90	3.00
59	CS603PC	Artificial Intelligence	90	0	0	0	90	3.00
60	CS632PE,CS633PE	Professional Elective-III	90	0	0	0	90	3.00
61	CE611OE,EE611OE	Open Elective-I	90	0	0	0	90	3.00
62	CS604PC	Machine Learning Laboratory	0	0	30	0	30	1.00
63	CS605PC	Artificial Intelligence Laboratory	0	0	30	0	30	1.00
64	CS642PE,CS643PE	Professional Electve-III laboratory	0	0	30	0	30	1.00
65	CS606PW	Industrial Oriented Mini Project/ Internship/Skill Development Course (Big data-Spark)	0	0	60	0	60	2.00
66	MS701HS	Management Science	90	0	0	0	90	3.00
67	CS702PC	Linux Programming	90	0	0	0	90	3.00
68	CS731PE,CS732PE	Professional Elective-III	90	0	0	0	90	3.00
69	CS742PE,CS743PE	Professional Elective-IV	90	0	0	0	90	3.00
70	CE721OE	Open Elective-II	90	0	0	0	90	3.00

		Total	**3180**	**270**	**930**	**420**	**4800**	**160.00**

1.2.3 Components of Curriculum (5) Institute Marks : 5.00

Course Components	Curriculum Content (% of total number of credits of the program)	Total number of contact hours	Total number of credits
Basic Sciences	14.06	675.00	22.50
Basic Engineering	13.44	645.00	21.50
Humanities and Social Scienc	6.25	300.00	10.00
Program Core	36.88	1770.00	59.00
Program Electives	12.50	600.00	20.00
Open Electives	5.63	270.00	9.00
Project(s)	10.00	480.00	16.00
Internships/Seminars	1.25	60.00	2.00
Any other (Please specify)	0.00	0.00	0.00
Total number of Credits			**160.00**

1.2.4 Strategies for Education Reforms (5) Institute Marks : 5.00

Strategies for Education Reforms

Anurag Engineering College (ANRK) has adopted a comprehensive strategy to implement education reforms in alignment with NEP 2020, focusing on curriculum innovation, multidisciplinary learning, flexible credit systems, and outcome-based assessment.

1. Curriculum Design & Multidisciplinary Approach

The institution follows a Choice Based Credit System (CBCS) (R18, R22, R25) that enables flexibility in curriculum design. The curriculum is periodically revised through Boards of Studies (BoS) and Academic Council to incorporate:

- Multidisciplinary and interdisciplinary courses
- Open electives across departments
- Emerging areas such as AI/ML, IoT, Cybersecurity, and Data Science
- Industry-oriented courses and skill-based modules

Students are encouraged to undertake minor/specialization tracks, promoting cross-disciplinary knowledge and holistic learning.

2. Academic Bank of Credits (ABC) & APAAR Integration

In line with UGC guidelines, the institution supports:

- Academic Bank of Credits (ABC): Students can accumulate and transfer credits earned from different recognized institutions.
- APAAR (Automated Permanent Academic Account Registry): Student academic records are digitally maintained for seamless mobility and verification.

Credits earned through SWAYAM / NPTEL / approved MOOCs are integrated into the curriculum, enabling flexible and lifelong learning pathways.

3. Outcome-Based Education (OBE) Framework

The institution follows a robust OBE framework, ensuring alignment of:

- Course Outcomes (COs)
- Program Outcomes (POs)
- Program Specific Outcomes (PSOs)

Assessment methods are mapped to outcomes, with CO–PO attainment analysis conducted regularly to improve teaching-learning processes.

4. Examination and Assessment Reforms

A major component of education reform is the transformation of the evaluation system from marks-based to competency-based assessment.

Key reforms implemented include:

- Continuous Internal Evaluation (CIE)
  - Multiple assessment tools: assignments, quizzes, seminars, lab performance, and viva
  - Two internal exams + provision for a third exam for fairness
  - Promotes continuous learning and reduces exam stress
- Semester End Examination (SEE)
  - Question papers designed using Bloom's Taxonomy
  - Balanced evaluation of LOTS and HOTS (analysis, design, problem-solving)
  - External examiners involved to ensure quality and transparency
- Digital Examination System
  - Use of Examination Management System (EMS) / BEES Examination Tool (BET) for:
    - Hall tickets, seating, coding/decoding
    - Result processing and grade sheet generation
  - Ensures efficiency, accuracy, and transparency
- Evaluation Transparency
  - Coding system for anonymity
  - Three-tier evaluation:
    - Initial evaluation
    - Moderation
    - Grievance redressal
  - Revaluation and photocopy provisions
- Rubric-Based & Experiential Assessment
  - Projects, internships, and mini-projects evaluated using rubrics
  - Focus on:
    - Innovation
    - Problem-solving
    - Teamwork
    - Industry relevance

5. Experiential & Skill-Based Learning

The institution emphasizes:

- Internships and industry projects
- Capstone and mini-projects
- Laboratory-based and activity-based learning
- Hackathons, innovation challenges, and startup incubation

This ensures students are assessed on real-world competencies and employability skills.

6. Digital & Data-Driven Academic Systems

- Implementation of LMS and ERP systems
- Digital content delivery and assessment
- Real-time tracking of attendance, performance, and outcomes
- Data-driven decision-making through academic audits

7. Continuous Improvement Mechanism

- Semester-wise academic audits and result analysis
- Feedback from students, faculty, and industry experts
- Faculty training on assessment design and OBE practices
- Periodic curriculum updates aligned with industry needs

The institution's education reform strategy integrates curriculum flexibility, digital systems, competency-based assessment, and multidisciplinary learning. Through the adoption of ABC, APAAR, OBE practices, and robust examination reforms, ANRK ensures a student-centric, transparent, and future-ready education system, aligned with NEP 2020 and global academic standards.

Data Validation & Verification Justification – Examination Reforms

1. Continuous Internal Evaluation (CIE):

The institution follows a structured CIE system with assignments, quizzes, seminars, lab performance, and internal exams. Records of internal marks and evaluation schemes are maintained.

2. Outcome-Based Assessment:

Question papers are designed using Bloom's Taxonomy, ensuring evaluation of higher-order skills (HOTS). CO–PO attainment reports and question papers serve as evidence.

3. Transparency in Evaluation:

- Coding and decoding of answer scripts
- Three-tier evaluation (evaluation, moderation, grievance redressal)
- Provision for revaluation and photocopy documentary proof includes examination manual and policy documents .

4. Digital Examination System:

The institution uses Examination Management System (EMS) / BEES Examination Tool (BET) for automation of examination processes. System reports and screenshots are available as evidence.

5. Rubric-Based & External Evaluation:

Projects, internships, and practicals are evaluated using rubrics, with involvement of external examiners. Evaluation sheets and rubrics are documented.

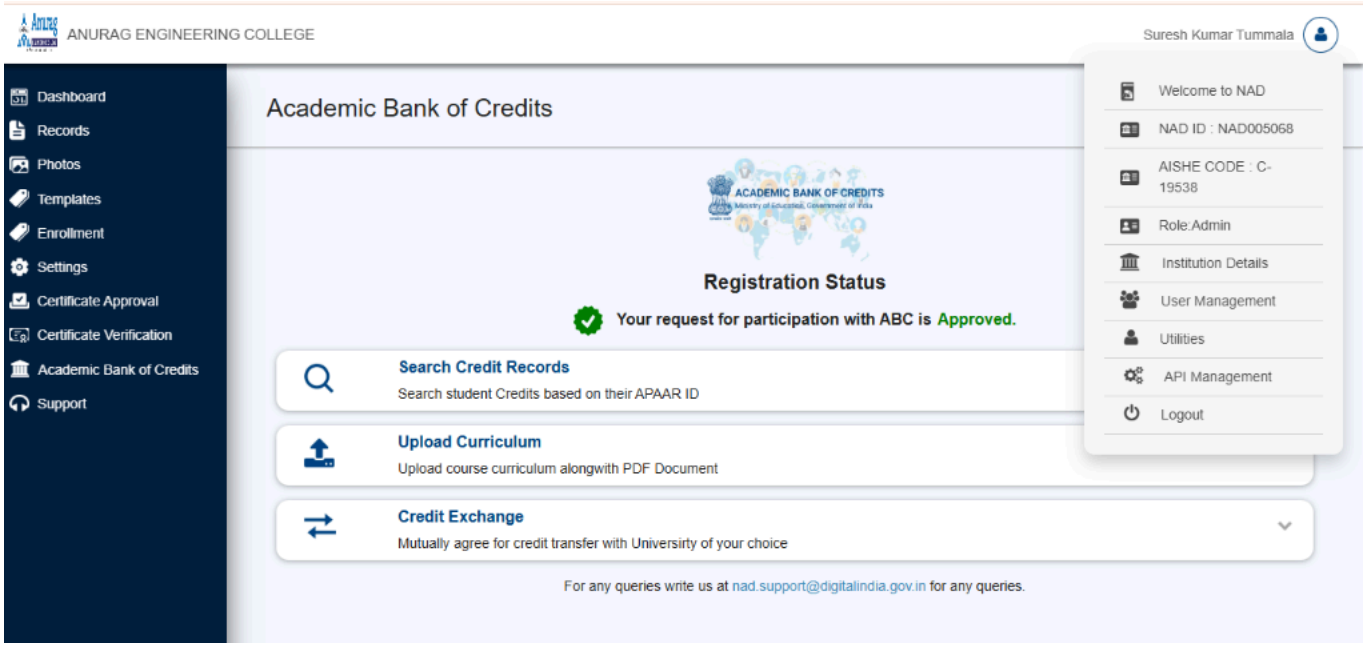
6. Grading System:

A relative grading system (SGPA/CGPA) is implemented as per CBCS regulations. Grade sheets and result processing documents are maintained.

7. Continuous Improvement:

Academic audits, result analysis, and stakeholder feedback are conducted periodically to improve assessment practices. IQAC reports and audit records serve as evidence.

All the above reforms are supported by documentary evidence such as academic regulations, curriculum structures, BoS/Academic Council minutes, examination manuals, student records, and digital system reports, which are made available for verification.



1.3 PO, PSO and their Mapping with Courses (20) — **Total Marks 20.00**

1.3.1 POs and PSOs (5) :

PSO1	

1.3.2 Mapping between the Courses and POs/PSOs (15) — Institute Marks : 15.00

PO:

PO Number	List of Courses
PO1	Matrices and Calculus, Engineering Chemistry, Applied Physics, Programming for Problem Solving, Python Programming, Probability and Statistics, Discrete Mathematics, Data Structures, Computer Organization and Architecture, Digital Electronics, Design and Analysis of Algorithms, Machine Learning, Artificial Intelligence
PO2	Programming for Problem Solving, Data Structures, Object Oriented Programming through Java, Operating Systems, Database Management Systems, Design and Analysis of Algorithms, Machine Learning, Artificial Intelligence
PO3	Software Engineering, Database Management Systems, Operating Systems, Computer Networks, DevOps, Compiler Design, Machine Learning, Artificial Intelligence, Real-time Research Project
PO4	Probability and Statistics, Machine Learning, Artificial Intelligence, Data Visualization (R/Power BI)
PO5	Python Programming Laboratory, Data Structures Laboratory, Java Laboratory, DBMS Laboratory, Operating Systems Laboratory, Computer Networks Laboratory, DevOps Laboratory, Machine Learning Laboratory, Artificial Intelligence Laboratory

PO6	Environmental Science, Business Economics & Financial Analysis, Computer Networks, Cryptography and Network Security, Real-time Research Project, Mini Project/Internship, Project Stage I, Project Stage II
PO7	Constitution of India, Intellectual Property Rights, Gender Sensitization, Environmental Science, Organizational Behavior
PO8	Engineering Workshop, IT Workshop, DevOps Laboratory, Computer Networks Laboratory, Mini Project/Internship, Project Stage I, Project Stage II
PO9	English for Skill Enhancement, English Communication Skills Laboratory, Advanced English Communication Skills Laboratory, Seminar, Real-time Research Project, Mini Project/Internship, Project Stage I, Project Stage II
PO10	Business Economics & Financial Analysis, Software Engineering, DevOps, Real-time Research Project, Mini Project/Internship, Project Stage I, Project Stage II
PO11	Artificial Intelligence, Machine Learning, Professional Electives, Open Electives, Skill Development Courses, , Real-time Research Project, Mini Project/Internship, Project Stage I, Project Stage II

PSO:

PO Number	List of Courses
PSO1	Matrices and Calculus, Probability and Statistics, Discrete Mathematics, Programming for Problem Solving, Python Programming, Data Structures, Object Oriented Programming through Java, Computer Organization and Architecture, Design and Analysis of Algorithms, Machine Learning, Artificial Intelligence.
PSO2	Software Engineering, Database Management Systems, Operating Systems, Computer Networks, DevOps, Cryptography and Network Security, Machine Learning, Artificial Intelligence, Mini Project/Internship, Real-time Research Project, Project Stage I, Project Stage II.

1.4 Course Outcomes and Course Articulation Matrix (30)

Total Marks 30.00

1.4.1 Course Outcome (Semester Wise) (15)

Institute Marks : 15.00

No. of Core Courses : 10	C1 :	C2 : 4	C3 : 4	C4 : 2	C5 :	C6 :	C7 :	C8 :

Note : Number of Outcomes for a Course is expected to be around 6.

Course Code :	CS303PC	Semester :	3

Course Outcome	Statements
CS303PC.1	Analyze the representation of various static, dynamic and, hierarchical data structures and Design and implement the mechanism of linear data structures
CS303PC.2	Outline the concepts of hashing, collision and its resolution methods using hash function
CS303PC.3	Design and Implementation of various advanced concepts of binary trees.
CS303PC.4	Implement various algorithms on graph data structures and implementation of various sorting techniques
CS303PC.5	Design and implementation of Pattern Matching algorithms to find patterns within a bigger set of data or text.

Course Code :	CS304PC	Semester :	3

Course Outcome	Statements

CS304PC.1	Understand the basics of instruction sets and their impact on processor design
CS304PC.2	Demonstrate an understanding of the design of the functional units of a digital computer system
CS304PC.3	Evaluate cost performance and design trade-offs in designing and constructing a computer processor including memory.
CS304PC.4	Design a pipeline for consistent execution of instructions with minimum hazards
CS304PC.5	Recognize and manipulate representations of numbers stored in digital computers

Course Code :	CS403PC	**Semester :**	4

Course Outcome	Statements
CS403PC.1	Will be able to control access to a computer and the files that may be shared
CS403PC.2	Demonstrate the knowledge of the components of computers and their respective roles in computing.
CS403PC.3	Ability to recognize and resolve user problems with standard operating environments
CS403PC.4	Gain practical knowledge of how programming languages, operating systems, and architectures interact and how to use each effectively.
CS403PC.5	Identify storage management and protection

Course Code :	CS404PC	**Semester :**	4

Course Outcome	Statements
CS404PC.1	Gain knowledge of fundamentals of DBMS and ER Model
CS404PC.2	Able to apply the knowledge of relational model and normalization
CS404PC.3	Apply the basics of SQL for retrieval and management of data
CS404PC.4	Be acquainted with the basics of transaction processing and concurrency control
CS404PC.5	Gain knowledge on database storage structures and access techniques

Course Code :	CS501PC	**Semester :**	5

Course Outcome	Statements
CS501PC.1	Acquire the knowledge of algorithm analysis and its notations that are applied on the problems solved by divide and conquer paradigm
CS501PC.2	Apply the major graph algorithms for model engineering problems and knowledge of the greedy paradigm
CS501PC.3	Apply the dynamic programming paradigm and recite algorithms that employ dynamic programming paradigm
CS501PC.4	Apply the concept of back tracking, branch and bound paradigm for real time problems
CS501PC.5	Analyse the complexity of problems and differentiate that in term of P and NP Problems

Course Code :	CS502PC	**Semester :**	5

Course Outcome	Statements
CS502PC.1	Analyze TCP/IP and OSI models and various protocols and various error handling mechanisms.

CS502PC.2	Use of various devices in connecting different types of LANs.
CS502PC.3	Analyze different routing protocols and IP Addresses
CS502PC.4	Discuss the various services offered by transport layer
CS502PC.5	Describes the working of various networked applications such as DNS, Mail, WWW and HTTP.

Course Code :	CS601PC	**Semester :**	6

Course Outcome	Statements
CS601PC.1	Demonstrate the foundation Machine Learning concepts
CS601PC.2	Generate models using Perception in real World scenarios
CS601PC.3	Build appropriate models for real World problem solving using Supervised Learning.
CS601PC.4	Interpret algorithm results and transform them into actionable information suitable for real-time applications in Unsupervised Learning.
CS601PC.5	Device Reinforcement Learning models

Course Code :	CS602PC	**Semester :**	6

Course Outcome	Statements
CS602PC.1	Gain proficiency in classifying machines by their power in recognizing languages.
CS602PC.2	Learn to employ finite state machines for modelling and solving computing problems
CS602PC.3	Design context free grammars for formal languages and master in context-free languages
CS602PC.4	Design context sensitive grammars for formal languages
CS602PC.5	Distinguish between decidability and undecidability

Course Code :	CS701PC	**Semester :**	7

Course Outcome	Statements
CS701PC.1	Understand basic cryptographic techniques, categories of attacks
CS701PC.2	Understand basic cryptographic algorithms, message and web authentication and security issues
CS701PC.3	Identify information system requirements for both of them such as client and server
CS701PC.4	Understand the current legal issues towards information security
CS701PC.5	Understand the Email Security and other security techniques

Course Code :	CS702PC	**Semester :**	7

Course Outcome	Statements
CS702PC.1	Understand the ability to design a compiler based on a set of language features, incorporating patterns, tokens, and regular expressions for lexical analysis
CS702PC.2	Develop scanner and implement LL and LR parsers by using the lex and yacc tools

CS702PC.3	Apply syntactic analysis techniques and generate intermediate code for different programs.
CS702PC.4	Design algorithms for code optimization to improve program performance in terms of space and time complexity
CS702PC.5	Develop algorithms to generate efficient machine code

1.4.2 Course Articulation Matrix (15) :

1 . course name : CS303PC

Course	Statements	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CS303PC.1	Analyze the	Select	Select	Select	Select	Select	Select	Select	Select	Select	Select	Select
CS303PC.2	Outline the c	Select	Select	Select	Select	Select	Select	Select	Select	Select	Select	Select
CS303PC.3	Design and	Select	Select	Select	Select	Select	Select	Select	Select	Select	Select	Select
CS303PC.4	Implement v	Select	Select	Select	Select	Select	Select	Select	Select	Select	Select	Select
CS303PC.5	Design and i	Select	Select	Select	Select	Select	Select	Select	Select	Select	Select	Select

2 . course name : CS304PC

Course	Statements	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CS304PC.1	Understand	Select	Select	Select	Select	Select	Select	Select	Select	Select	Select	Select
CS304PC.2	Demonstrate	Select	Select	Select	Select	Select	Select	Select	Select	Select	Select	Select
CS304PC.3	Evaluate cos	Select	Select	Select	Select	Select	Select	Select	Select	Select	Select	Select
CS304PC.4	Design a pip	Select	Select	Select	Select	Select	Select	Select	Select	Select	Select	Select
CS304PC.5	Recognize a	Select	Select	Select	Select	Select	Select	Select	Select	Select	Select	Select

3 . course name : CS403PC

Course	Statements	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CS403PC.1	Will be able	Select	Select	Select	Select	Select	Select	Select	Select	Select	Select	Select
CS403PC.2	Demonstrate	Select	Select	Select	Select	Select	Select	Select	Select	Select	Select	Select
CS403PC.3	Ability to rec	Select	Select	Select	Select	Select	Select	Select	Select	Select	Select	Select
CS403PC.4	Gain practic	Select	Select	Select	Select	Select	Select	Select	Select	Select	Select	Select
CS403PC.5	Identify stora	Select	Select	Select	Select	Select	Select	Select	Select	Select	Select	Select

4 . course name : CS404PC

Course	Statements	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CS404PC.1	Gain knowle	Select	Select	Select	Select	Select	Select	Select	Select	Select	Select	Select

CS404PC.2	Able to appl	Select	Select	Select	Select	Select	Select	Select	Select	Select	Select	Select
CS404PC.3	Apply the ba	Select	Select	Select	Select	Select	Select	Select	Select	Select	Select	Select
CS404PC.4	Be acquaint	Select	Select	Select	Select	Select	Select	Select	Select	Select	Select	Select
CS404PC.5	Gain knowle	Select	Select	Select	Select	Select	Select	Select	Select	Select	Select	Select

5 . course name : CS501PC

Course	Statements	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CS501PC.1	Acquire the	Select	Select	Select	Select	Select	Select	Select	Select	Select	Select	Select
CS501PC.2	Apply the m	Select	Select	Select	Select	Select	Select	Select	Select	Select	Select	Select
CS501PC.3	Apply the dy	Select	Select	Select	Select	Select	Select	Select	Select	Select	Select	Select
CS501PC.4	Apply the cc	Select	Select	Select	Select	Select	Select	Select	Select	Select	Select	Select
CS501PC.5	Analyse the	Select	Select	Select	Select	Select	Select	Select	Select	Select	Select	Select

6 . course name : CS502PC

Course	Statements	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CS502PC.1	Analyze TCI	Select	Select	Select	Select	Select	Select	Select	Select	Select	Select	Select
CS502PC.2	Use of variou	Select	Select	Select	Select	Select	Select	Select	Select	Select	Select	Select
CS502PC.3	Analyze diffe	Select	Select	Select	Select	Select	Select	Select	Select	Select	Select	Select
CS502PC.4	Discuss the	Select	Select	Select	Select	Select	Select	Select	Select	Select	Select	Select
CS502PC.5	Describes th	Select	Select	Select	Select	Select	Select	Select	Select	Select	Select	Select

7 . course name : CS601PC

Course	Statements	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CS601PC.1	Demonstrate	Select	Select	Select	Select	Select	Select	Select	Select	Select	Select	Select
CS601PC.2	Generate m	Select	Select	Select	Select	Select	Select	Select	Select	Select	Select	Select
CS601PC.3	Build approp	Select	Select	Select	Select	Select	Select	Select	Select	Select	Select	Select
CS601PC.4	Interpret alg	Select	Select	Select	Select	Select	Select	Select	Select	Select	Select	Select
CS601PC.5	Device Reinf	Select	Select	Select	Select	Select	Select	Select	Select	Select	Select	Select

8 . course name : CS602PC

Course	Statements	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CS602PC.1	Gain proficie	Select	Select	Select	Select	Select	Select	Select	Select	Select	Select	Select
CS602PC.2	Learn to em	Select	Select	Select	Select	Select	Select	Select	Select	Select	Select	Select
CS602PC.3	Design cont	Select	Select	Select	Select	Select	Select	Select	Select	Select	Select	Select
CS602PC.4	Design cont	Select	Select	Select	Select	Select	Select	Select	Select	Select	Select	Select

CS602PC.5	Distinguish I	Select	Select	Select	Select	Select	Select	Select	Select	Select	Select	Select

9 . course name : CS701PC

Course	Statements	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CS701PC.1	Understand	Select	Select	Select	Select	Select	Select	Select	Select	Select	Select	Select
CS701PC.2	Understand	Select	Select	Select	Select	Select	Select	Select	Select	Select	Select	Select
CS701PC.3	Identify infor	Select	Select	Select	Select	Select	Select	Select	Select	Select	Select	Select
CS701PC.4	Understand	Select	Select	Select	Select	Select	Select	Select	Select	Select	Select	Select
CS701PC.5	Understand	Select	Select	Select	Select	Select	Select	Select	Select	Select	Select	Select

10 . course name : CS702PC

Course	Statements	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CS702PC.1	Understand	Select	Select	Select	Select	Select	Select	Select	Select	Select	Select	Select
CS702PC.2	Develop sca	Select	Select	Select	Select	Select	Select	Select	Select	Select	Select	Select
CS702PC.3	Apply synta	Select	Select	Select	Select	Select	Select	Select	Select	Select	Select	Select
CS702PC.4	Design algo	Select	Select	Select	Select	Select	Select	Select	Select	Select	Select	Select
CS702PC.5	Develop algo	Select	Select	Select	Select	Select	Select	Select	Select	Select	Select	Select

1 . Course Name : CS303PC

Course	PSO1	PSO2
CS303PC.1	Select	Select
CS303PC.2	Select	Select
CS303PC.3	Select	Select
CS303PC.4	Select	Select
CS303PC.5	Select	Select

2 . Course Name : CS304PC

Course	PSO1	PSO2
CS304PC.1	Select	Select
CS304PC.2	Select	Select
CS304PC.3	Select	Select
CS304PC.4	Select	Select
CS304PC.5	Select	Select

3 . Course Name : CS403PC

Course	PSO1	PSO2
CS403PC.1	Select	Select
CS403PC.2	Select	Select
CS403PC.3	Select	Select
CS403PC.4	Select	Select
CS403PC.5	Select	Select

4 . Course Name : CS404PC

Course	PSO1	PSO2
CS404PC.1	Select	Select
CS404PC.2	Select	Select
CS404PC.3	Select	Select
CS404PC.4	Select	Select
CS404PC.5	Select	Select

5 . Course Name : CS501PC

Course	PSO1	PSO2
CS501PC.1	Select	Select
CS501PC.2	Select	Select
CS501PC.3	Select	Select
CS501PC.4	Select	Select
CS501PC.5	Select	Select

6 . Course Name : CS502PC

Course	PSO1	PSO2
CS502PC.1	Select	Select
CS502PC.2	Select	Select
CS502PC.3	Select	Select
CS502PC.4	Select	Select
CS502PC.5	Select	Select

7 . Course Name : CS601PC

Course	PSO1	PSO2
CS601PC.1	Select	Select
CS601PC.2	Select	Select

CS601PC.3	Select	Select
CS601PC.4	Select	Select
CS601PC.5	Select	Select

8 . Course Name : CS602PC

Course	PSO1	PSO2
CS602PC.1	Select	Select
CS602PC.2	Select	Select
CS602PC.3	Select	Select
CS602PC.4	Select	Select
CS602PC.5	Select	Select

9 . Course Name : CS701PC

Course	PSO1	PSO2
CS701PC.1	Select	Select
CS701PC.2	Select	Select
CS701PC.3	Select	Select
CS701PC.4	Select	Select
CS701PC.5	Select	Select

10 . Course Name : CS702PC

Course	PSO1	PSO2
CS702PC.1	Select	Select
CS702PC.2	Select	Select
CS702PC.3	Select	Select
CS702PC.4	Select	Select
CS702PC.5	Select	Select

1.5 Program Articulation Matrix (5) — **Total Marks 5.00**

Program Articulation Matrix :

Course Code	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
MA101BS	3.00	2.20	1.00	1.40	1.00	0.00	0.00	0.00	0.00	0.00	1.00
CH102BS	2.33	2.66	2.20	1.50	1.66	0.00	0.00	0.00	0.00	0.00	0.00
CS103ES	3.00	2.4	2.00	0.00	2.00	0.00	0.00	0.00	0.00	0.00	0.00
EE104ES	3.00	2.80	2.40	1.80	1.20	2.33	0.00	0.00	0.00	0.00	0.00
EG105ES	1.25	2.00	2.5	2.33	1.00	0.00	0.00	0.00	0.00	2.00	0.00
CS106ES	2.40	2.40	1.40	1.50	1.80	1.00	2.00	0.00	0.00	0.00	1.20
CS107BS	2.00	2.00	2.00	1.50	1.67	2.33	1.33	1.00	1.00	2.00	1.00
CS109ES	1.60	2.40	2.00	2.20	0.00	0.00	0.00	0.00	0.00	0.00	0.00
EE108ES	3.00	2.80	1.60	3.00	2.00	0.00	0.00	0.00	0.00	0.00	1.00
HS110MC	0.00	0.00	0.00	0.00	0.00	2.20	2.25	1.00	1.00	0.00	1.00
MA201BS	3.00	2.40	1.60	1.80	1.00	0.00	0.00	0.00	0.00	0.00	1.00
AP202BS	1.80	1.75	1.00	1.25	0.00	0.00	0.00	0.00	0.00	0.00	0.00
ME203ES	2.25	1.00	0.00	1.00	1.50	0.00	0.00	0.00	2.00	0.00	1.00
EN204HS	0.00	0.00	0.00	0.00	0.00	0.00	2.00	2.00	1.40	2.60	1.60
EC205ES	3.00	1.80	2.60	0.00	0.00	1.60	1.00	0.00	0.00	0.00	0.00
AP206BS	1.75	1.25	0.00	0.00	0.00	0.00	0.00	0.00	1.75	1.00	0.00
CS207ES	1.60	2.20	2.40	2.20	3.00	0.00	0.00	0.00	0.00	0.00	0.00
EN208HS	0.00	0.00	0.00	0.00	0.00	0.00	1.60	2.20	2.60	1.80	2.40
CS209ES	1.75	2.25	1.00	1.00	2.00	1.00	1.00	1.50	2.00	1.00	1.00
ES210MC	2.00	2.00	0.80	0.00	0.00	2.40	2.80	2.80	0.00	0.00	1.40
PS301BS	2.80	2.40	1.60	2.00	1.80	1.00	0.00	0.00	0.00	0.00	2.20
DE302ES	2.40	2.60	3.00	1.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
CS303PC	3.00	2.80	2.00	2.00	2.20	0.00	0.00	1.00	0.00	0.00	1.80
CS304PC	2.80	2.40	1.60	1.00	1.80	1.00	0.00	0.00	0.00	1.00	1.20
CS305PC	2.40	2.40	2.40	1.00	2.60	1.00	1.00	1.00	1.00	1.00	1.80
CS306PC	3.00	2.80	2.40	0.00	0.00	0.00	0.00	1.00	0.00	0.00	2.20
CS307PC	2.20	2.60	2.20	1.00	2.40	1.00	1.00	1.00	1.20	1.00	2.00
HS309MC	0.00	0.00	0.00	0.00	0.00	2.60	2.60	2.00	2.00	0.00	0.00
CS308PC	2.00	2.00	2.40	2.40	2.40	2.00	1.40	2.00	2.20	1.80	2.20
MB401HS	2.40	2.60	1.40	1.67	1.00	1.80	1.20	1.00	1.20	2.40	2.00
CS402PC	3.00	2.80	1.60	2.20	1.80	0.00	0.00	0.00	0.00	0.00	1.00

CS403PC	2.60	2.00	1.25	1.00	1.60	0.00	0.00	1.00	1.00	0.00	1.25
CS404PC	2.80	2.40	2.20	1.40	2.00	0.00	1.00	0.00	1.00	0.00	1.20
CS405PC	2.80	2.60	1.80	1.40	2.40	1.50	1.60	2.00	0.00	0.00	1.67
CS406PC	3.00	2.20	2.00	1.75	1.33	0.00	0.00	1.00	1.00	0.00	0.00
CS407PC	3.00	2.60	2.40	1.40	2.20	0.00	0.00	1.00	1.00	0.00	1.20
CS408PW	1.67	2.20	1.40	1.60	2.00	1.80	1.67	1.50	1.60	1.67	2.20
HS411MC	0.00	1.40	1.00	0.00	0.00	2.20	3.00	1.00	1.00	0.00	1.20
CS409PC	3.00	2.60	2.80	2.20	3.00	1.60	1.00	2.00	3.00	1.40	2.00
CS501PC	3.00	2.80	2.00	1.80	2.80	0.00	0.00	0.00	0.00	0.00	2.40
CS502PC	2.60	2.40	1.00	1.50	2.20	0.00	0.00	0.00	1.00	0.00	1.00
CS503PC	2.40	2.20	2.00	1.00	2.00	1.00	1.00	1.00	1.00	3.00	1.20
CS513PE	2.67	2.33	2.67	2.75	2.50	3.00	1.00	2.00	3.00	2.00	1.00
CS515PE	3.00	2.25	2.40	1.80	2.50	1.00	1.00	2.00	1.25	1.00	1.00
CS523PE	2.60	2.40	1.00	1.50	2.20	0.00	0.00	0.00	1.00	0.00	1.00
CS524PE	2.60	2.40	2.40	2.25	2.25	0.00	0.00	0.00	0.00	0.00	0.00
CS504PC	2.40	2.40	2.00	1.40	2.00	1.00	2.00	1.00	2.00	1.20	1.00
CS505PC	2.60	2.00	2.00	1.60	2.60	0.00	0.00	2.00	2.00	3.00	0.00
AE506HS	0.00	0.00	0.00	0.00	0.00	0.00	1.60	2.20	2.60	1.80	2.40
CS507PC	2.60	2.80	2.50	2.00	2.00	1.00	0.00	0.00	0.00	0.00	2.20
CS601PC	3.00	2.80	2.40	2.20	2.00	1.00	0.00	0.00	0.00	0.00	2.00
CS602PC	2.80	3.00	2.40	2.20	2.00	0.00	0.00	2.00	0.00	0.00	0.00
CS603PC	2.40	2.20	1.75	1.80	2.40	0.00	0.00	1.00	0.00	0.00	2.20
CS632PE	1.00	1.40	2.00	2.00	1.00	2.00	0.00	0.00	2.00	2.00	1.40
CS633PE	2.60	2.40	2.00	1.80	2.20	3.00	0.00	0.00	0.00	2.00	0.00
CE611OE	1.40	3.00	1.67	2.00	2.50	1.33	2.67	1.00	2.00	1.67	2.00
EE611OE	2.60	2.25	3.00	2.00	2.00	2.00	1.00	0.00	0.00	1.50	1.80
CS604PC	2.80	2.80	2.00	2.60	2.60	1.00	0.00	2.00	2.00	1.00	1.60
CS605PC	2.80	2.40	2.00	1.67	2.20	1.00	1.00	2.00	2.00	1.50	1.40
CS642PE	1.00	2.00	1.00	0.00	1.00	1.00	0.00	0.00	2.00	2.00	2.00
CS643PE	2.80	2.80	2.40	2.80	2.80	2.00	0.00	2.00	2.00	2.00	0.00
CS606PW	0.80	2.00	1.60	1.20	1.20	1.20	1.00	1.40	1.60	2.00	2.00
MS701HS	2.67	2.33	2.67	2.75	2.00	3.00	1.00	2.00	3.00	2.00	1.00

CS731PE	2.60	2.60	2.00	1.67	2.60	1.00	0.00	0.00	1.00	0.00	1.20
CS732PE	1.80	2.20	2.67	2.25	3.00	2.00	0.00	0.00	0.00	2.00	0.00
CS742PE	2.67	2.33	2.67	2.75	2.00	3.00	1.00	2.00	3.00	2.00	1.00
CS743PE	2.20	2.20	1.75	1.60	2.20	1.00	0.00	1.50	1.00	1.00	2.20
CS721OE	1.80	1.50	1.50	2.00	1.75	1.75	1.00	1.67	2.00	1.75	1.50
CS703PC	1.20	1.80	1.60	2.00	2.00	0.00	0.00	0.00	0.00	0.00	0.00
CS704PW	.80	2.20	1.60	1.20	1.20	1.20	1.00	1.40	1.60	1.80	1.80
CS705PW	2.00	1.00	2.50	1.50	1.50	1.00	2.00	1.00	2.00	2.00	1.50
CS852PE	1.80	2.20	2.67	2.25	2.00	0.00	0.00	0.00	0.00	2.00	0.00
CS853PE	1.80	2.50	1.67	2.00	3.00	0.00	0.00	1.00	2.50	1.50	2.40
CS861PE	2.67	2.33	2.67	2.75	2.00	3.00	1.00	2.00	3.00	2.00	1.00
CS862PE	2.67	2.33	2.67	2.75	2.00	3.00	1.00	2.00	3.00	2.00	1.00
CE831OE	1.50	1.75	2.00	1.50	1.60	2.00	1.00	1.33	2.00	1.00	1.75
CE832OE	1.00	2.00	2.67	1.75	1.33	1.50	1.00	1.67	2.00	1.67	1.33
CS801PW	3.00	2.80	2.75	3.00	3.00	2.00	0.00	3.00	3.00	2.33	3.00
CS802PW	1.50	1.67	2.00	2.00	2.00	1.50	2.00	3.00	1.00	1.00	2.00
CS702PC	1.20	2.00	1.80	1.40	2.00	0.00	0.00	0.00	0.00	0.00	0.00

Course Code	PSO1	PSO2
AE506HS	0.00	1.60
AP202BS	1.66	0.00
AP206BS	0.00	1.33
CE611OE	2.25	2.50
CE721OE	1.80	2.20
CE831OE	2.25	2.50
CE832OE	2.40	2.50
CH102BS	1.5	2.00
CH107BS	2.5	2.5
CS103ES	3.00	1.80
CS106ES	2.40	1.80
CS109ES	3.00	1.40
CS207ES	2.40	2.20
CS209ES	1.75	1.40

CS303PC	2.40	2.60
CS304PC	2.80	2.20
CS305PC	2.40	2.40
CS306PC	2.80	2.40
CS307PC	2.60	2.20
CS308PC	2.60	2.60
CS402PC	2.00	1.60
CS403PC	2.20	2.40
CS404PC	2.60	2.20
CS405PC	2.20	2.60
CS406PC	2.60	2.20
CS407PC	3.00	1.60
CS408PW	2.40	2.80
CS409PC	3.00	2.80
CS501PC	2.60	2.20
CS502PC	2.40	2.20
CS503PC	2.20	3.00
CS504PC	2.50	2.25
CS505PC	2.40	3.00
CS507PC	2.80	3.00
CS513PE	2.40	2.20
CS515PE	2.25	2.25
CS523PE	2.40	2.20
CS524PE	2.80	2.60
CS601PC	2.60	2.20
CS602PC	2.60	2.60
CS603PC	2.80	2.60
CS604PC	2.60	2.00
CS605PC	2.60	2.00
CS606PW	2.80	2.40
CS610MC	3.00	3.00
CS632PE	2.40	2.20

CS633PE	2.20	2.40
CS642PE	2.40	2.20
CS643PE	2.80	2.40
CS702PC	2.40	2.20
CS703PC	2.40	2.20
CS704PW	2.66	3.00
CS705PW	2.20	2.40
CS731PE	2.40	2.40
CS732PE	2.00	2.80
CS742PE	2.00	3.00
CS743PE	2.60	2.20
CS801PW	2.50	3.00
CS802PW	2.80	2.60
CS852PE	2.40	3.00
CS853PE	2.20	2.80
CS861PE	2.00	2.20
CS862PE	2.00	3.00
DE302ES	3.00	2.00
EE104ES	3.00	3.00
EE108ES	2.00	1.50
EE611OE	2.20	2.25
EG105ES	1.00	1.50
EN204HS	0.00	1.80
EN208HS	0.00	1.60
ES205ES	1.00	2.60
ES210MC	0.00	1.20
HS110MC	0.00	1.00
HS309MC	0.00	2.20
HS411MC	0.00	1.80
MA101BS	1.60	1.00
MA201BS	1.60	1.00
MB401HS	1.50	2.00

ME203ES	0.00	1.00
MS701HS	2.00	3.00
PS301BS	1.80	1.60